

Joshua Oti

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EDUCATION

University of Southern California **Los Angeles, CA**
Master of Science, Computer Science; **GPA: 3.76/4.0** May 2026

- **Relevant Courses:** Machine Learning, NLP, Algorithms, Database Systems, Software Engineering

Covenant University **Ogun, Nigeria**
Bachelor of Engineering, Mechanical Engineering; **GPA: 4.85/5.0** Oct 2021

- **Relevant Courses:** Advanced Calculus, Linear Algebra, Probability & Statistics, Numerical Methods
- **Academic Honors:** Top 1% of Graduating Class (Mechanical Engineering); 3rd Place, ACPC (Regional ICPC)

EXPERIENCE

Autodesk **San Francisco, CA**
Machine Learning Engineer Intern May 2025 - Aug 2025

- Implemented few-shot prompting strategies to significantly enhance generative model reasoning and image output quality.
- Engineered a responsive Vue.js gamified app with enterprise SSO to drive engagement with the Autodesk culture code, resulting in a 20% increase in participation through seamless, device-agnostic accessibility.
- Owned the end-to-end lifecycle of a production FastAPI backend leveraging Azure OpenAI and Amazon S3 to architect a dynamic LLM pipeline; saved 10+ hours weekly by automating personalized image and text generation.

Phasecurve Technologies Ltd **Lagos, Nigeria**
Software Engineer Nov 2022 - Sep 2024

- Optimized content uploads in the Flask backend with multi-threading, reducing response time by 80%.
- Architected and maintained three Flask-based microservices using Azure SQL + Storage Account, successfully scaling the Tiddix project to handle 300% more traffic.
- Automated CI/CD pipelines with GitHub Actions and Docker, containerizing services and cutting deployment time by 50%.
- Led migration from Mailgun API to Gmail API, optimizing email service and reducing operational costs for the team.

ReHboX **Lagos, Nigeria**
Machine Learning Engineer Jun 2021 - Feb 2022

- Developed a real-time algorithm for edge devices to analyze human pose from camera inputs using OpenCV, boosting team productivity by 20% and enabling timely product delivery.
- Built a predictive model using Scikit-learn and an interactive dashboard to analyze survey data, providing insights on the total addressable market (TAM).
- Coauthored 6 design documents, aligning user stories to cut down meeting time by 15%.

SELECTED PROJECTS

LLM Fine-Tuning & Bias Analysis | PyTorch, Transformers — NLP Research **Nov 2025**

- Benchmarked bias shifts across decoder (Llama 3.2-3B) and encoder (RoBERTa/DeBERTa) architectures using StereoSet metrics (ICAT, LMS, SS) to quantify the trade-off between sentiment accuracy and social safety.
- Engineered a custom LoRA adapter targeting 7 projection layers (q, k, v, o, gate, up, down) with just 0.18% trainable parameters, enabling efficient training on A100 GPUs while maintaining comparable accuracy to full fine-tuning.
- Discovered that 8-shot in-context learning improved language modeling consistency (LMS) by 5.2 points without altering static bias, while nonsense-prompts paradoxically maximized safety scores (ICAT 73.79) over anti-stereotypical inputs.

Statistical Machine Learning Portfolio | Scikit-Learn, PyTorch — Advanced ML Modeling **Jan 2025**

- Architected active learning and semi-supervised self-training pipelines using L_1 -penalized SVMs and Monte Carlo simulations, benchmarking label efficiency against unsupervised approaches like Spectral Clustering.
- Developed predictive ensemble systems (Random Forests, XGBoost) on highly imbalanced engineering datasets, integrating SMOTE oversampling and cost-complexity pruning to prevent overfitting and enhance model interpretability.
- Engineered time-domain feature extraction frameworks for multi-sensor time series data and built regularized regression models (LASSO, Ridge, PCR) validated via stratified cross-validation.

Movies Dataset Visualization | Python, React, Ollama — Full-Stack & GenAI **Nov 2024**

- Engineered an agentic web application with a React frontend to enable natural-language querying and interactive data analysis over a large-scale movie dataset.
- Architected a hybrid search pipeline combining structured SQL queries (SQLite) with semantic vector retrieval (ChromaDB) to bridge complex analytical insights and semantic lookups, orchestrated by DeepSeek-R1 and Qwen3 LLMs.
- Validated the hybrid generation quality and agentic decision-making processes using the RAGAS evaluation framework to ensure factually accurate responses.

SKILLS & INTERESTS

- **Languages:** Python, C, C++, TypeScript, SQL, Java, C#, bash
- **AI/ML:** PyTorch, LangChain, MCP, Transformers, LoRA, Scikit-learn, ChromaDB, RAGAS, Google AI Studio
- **Tools:** Git, RabbitMQ, Kafka, Figma, GraphQL, Grafana, Prometheus, Miro, Postman, JMeter, MLflow
- **Cloud/Infra:** Microsoft Azure, Amazon Web Services, Google Cloud, Docker, CI/CD
- **Certifications:** AWS Certified Solutions Architect Associate (AWS CSAA)
- **Web/Backend:** FastAPI, gRPC, React, Vue.js, Flask, Node.js, Spring Boot, ASP.NET Core, Django, OAuth 2.0

LEADERSHIP

- **National Society of Black Engineers, Graduate Student Liaison USC Chapter** **Apr 2025 - Apr 2026**
Bridged the undergraduate-graduate gap by organizing joint social initiatives, driving a 25% increase in graduate student event attendance.